

Marketing Campaign Performance Analysis

1. Introduction

This project analyzes the performance of a bank marketing campaign to understand how different factors influence customer responses. The main objective is to identify which contact methods, months, weekdays, and previous campaign outcomes are associated with higher conversion rates.

The analysis was conducted using Python, Pandas, and Matplotlib. Microsoft Excel was also used to organize and examine the dataset.

2. Dataset Description

The dataset contains 41,188 customer records from a bank marketing campaign.

The target variable is **y**, which indicates whether a customer subscribed to a term deposit.

The two possible responses are:

- **Yes:** The customer subscribed.
- **No:** The customer did not subscribe.

The analysis focuses on five areas:

1. Overall campaign response rate.
2. Conversion rate by contact type.
3. Conversion rate by month.
4. Conversion rate by day of the week.
5. Conversion rate by previous campaign outcome.

3. Methodology

The dataset was analyzed using Python in Visual Studio Code.

Pandas was used to calculate customer response percentages and compare conversion rates across different categories.

Matplotlib was used to create charts that visually represent the results.

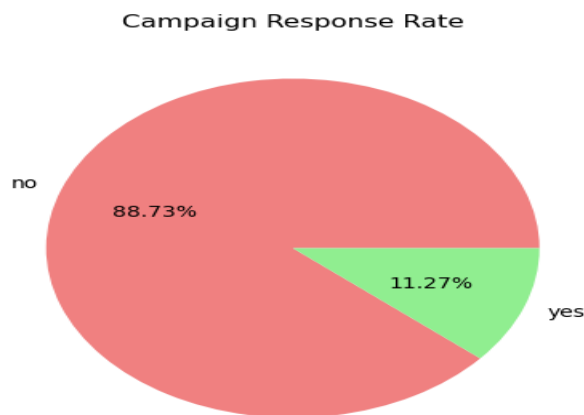
The conversion rate was calculated as follows:

Conversion Rate (%) = (Number of Yes Responses / Total Number of Responses in the Category) × 100

Five charts were created and saved as PNG files.

4. Results and Analysis

4.1 Overall Campaign Response Rate



The dataset contains 41,188 customer responses.

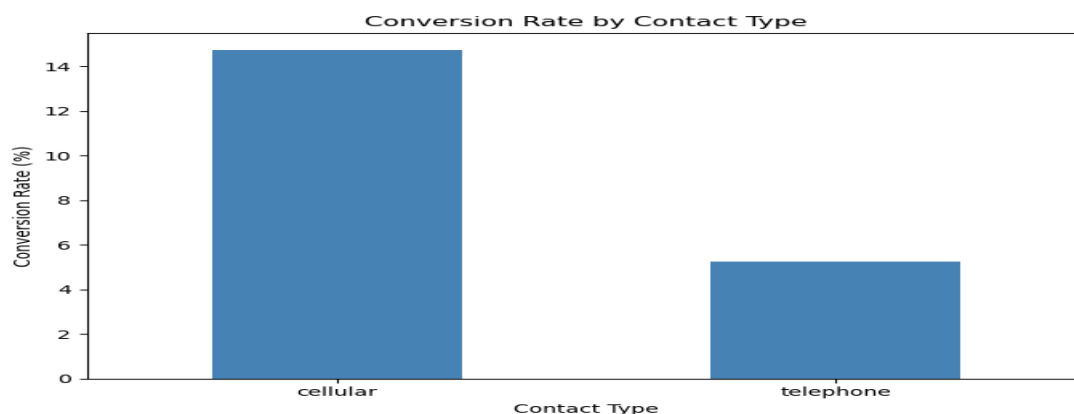
- No: 36,548 customers, representing 88.73%.
- Yes: 4,640 customers, representing 11.27%.

The overall conversion rate was **11.27%**.

This indicates that most customers did not subscribe to the term deposit. The campaign achieved a relatively low overall response rate, suggesting opportunities to improve customer targeting and campaign effectiveness.

Figure 1: Campaign Response Rate

4.2 Conversion Rate by Contact Type



The analysis compares two contact methods: cellular and telephone.

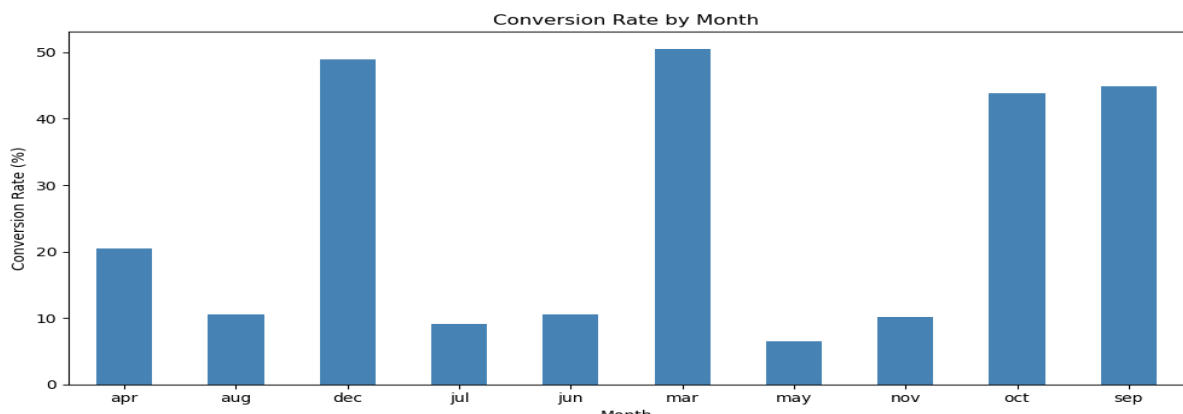
- Cellular: 14.74%.
- Telephone: 5.23%.

Cellular contact achieved a substantially higher conversion rate than telephone contact.

This suggests that cellular communication was more effective in generating positive customer responses during the campaign.

Figure 2: Conversion Rate by Contact Type

4.3 Conversion Rate by Month



Conversion rates varied considerably across different months.

March recorded the highest monthly conversion rate at approximately **50.55%**.

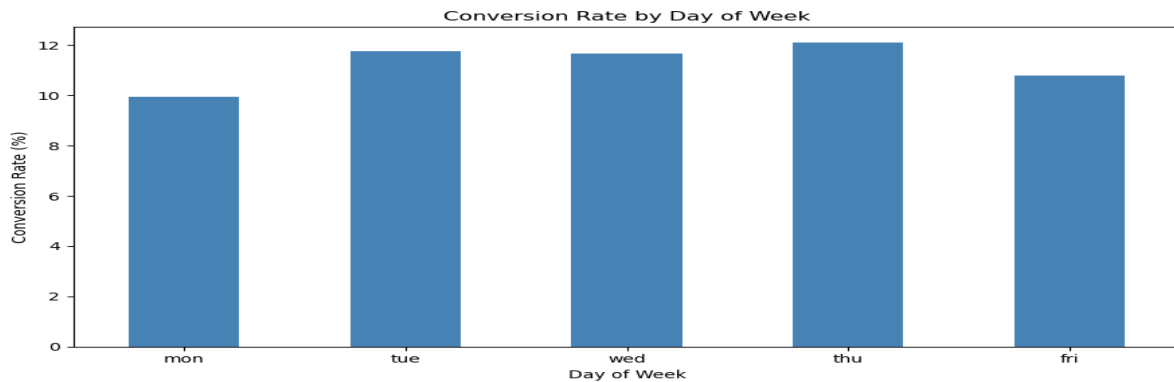
December, September, and October also showed relatively high conversion rates compared with several other months.

May and July recorded comparatively low conversion rates.

These results indicate that campaign performance differed depending on the month. However, monthly differences may also reflect changes in customer characteristics, campaign conditions, and the number of customers contacted.

Figure 3: Conversion Rate by Month

4.4 Conversion Rate by Day of the Week



The conversion rates for different weekdays were:

Day	Conversion Rate
Monday	9.95%
Tuesday	11.78%
Wednesday	11.67%
Thursday	12.12%
Friday	10.81%

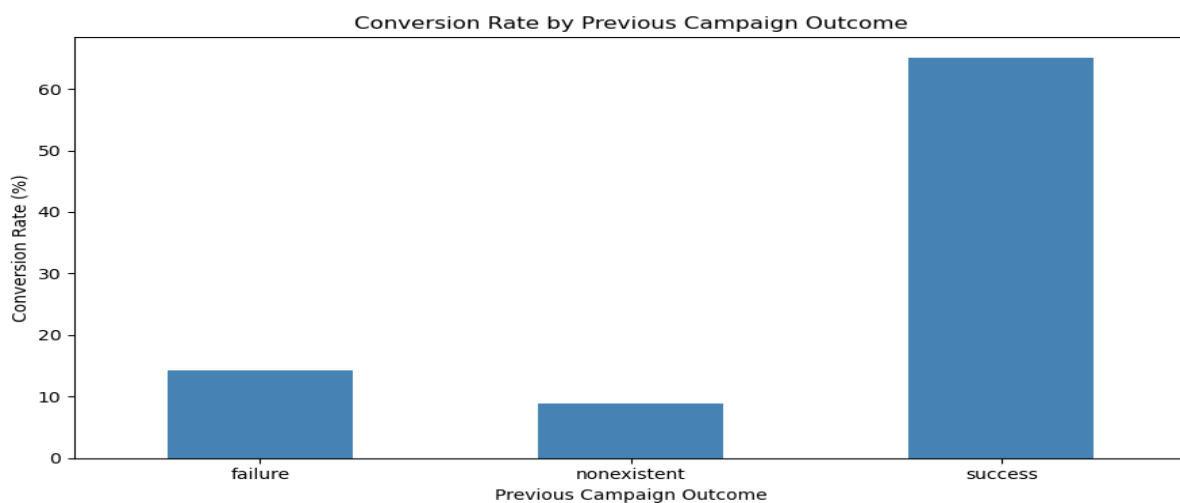
Thursday achieved the highest conversion rate at **12.12%**.

Monday recorded the lowest conversion rate at **9.95%**.

Although the differences between weekdays were relatively small, Thursday performed slightly better than the other days.

Figure 4: Conversion Rate by Day of Week

4.5 Conversion Rate by Previous Campaign Outcome



The analysis examined whether the outcome of a previous marketing campaign was associated with the customer's response to the current campaign.

The results were:

Previous Campaign Outcome	Conversion Rate
Failure	14.23%
Nonexistent	8.83%
Success	65.11%

Customers whose previous campaign outcome was successful had the highest conversion rate at **65.11%**.

Customers with no previous campaign outcome recorded had the lowest conversion rate at **8.83%**.

This indicates a strong association between previous campaign success and a positive response in the current campaign.

Figure 5: Conversion Rate by Previous Campaign Outcome

5. Key Findings

The analysis identified five main findings.

1. The overall campaign conversion rate was 11.27%, meaning that most contacted customers did not subscribe.
2. Cellular contact was associated with a higher conversion rate than telephone contact.
3. March recorded the highest monthly conversion rate, at approximately 50.55%.
4. Thursday had the highest conversion rate among weekdays, at 12.12%.
5. Previous campaign success was strongly associated with positive customer responses, with a conversion rate of 65.11%.

6. Recommendations

Based on the analysis, the following recommendations are proposed.

Improve customer targeting. Customers with successful previous campaign outcomes may be a valuable segment for future marketing activities.

Prioritize effective contact channels. Cellular communication should receive greater attention because it achieved a higher conversion rate in the analyzed data.

Evaluate campaign timing. The bank should investigate why some months achieved much higher conversion rates and consider these patterns when planning future campaigns.

Optimize weekday scheduling. Thursday may be considered for additional campaign activity, although the relatively small differences between weekdays should be taken into account.

Conduct further analysis. Additional variables, including age, occupation, education, and economic conditions, could help explain differences in campaign performance.

These recommendations are based on observed associations and do not establish that changing contact methods or campaign timing would necessarily cause higher conversion rates.

7. Conclusion

The Marketing Campaign Performance Analysis project examined 41,188 customer records to evaluate the effectiveness of a bank marketing campaign.

The overall conversion rate was 11.27%.

The results showed that cellular contact, certain months, and previous campaign success were associated with higher conversion rates.

Among the factors examined, previous campaign outcome showed a particularly large difference in customer responses.

The findings can support future marketing decisions by helping identify customer segments and campaign characteristics associated with successful outcomes.

Overall, the project demonstrates how data analysis and visualization can help organizations evaluate marketing performance and make more informed decisions.

8. Tools and Technologies

- Python
- Pandas
- Matplotlib
- Visual Studio Code
- Microsoft Excel
- Microsoft Word

9. Project Files

The project includes:

- `marketing_analysis.py`, containing the Python analysis.
- The original Excel dataset.
- Five PNG charts showing the analysis results.
- This written report.

10. Data Source

UCI Machine Learning Repository.

Bank Marketing Dataset, Bank Additional Full.

The dataset contains information about direct marketing campaigns conducted by a Portuguese banking institution.